



Goals vs. Actions as User-Facing Representations for Robot Programming

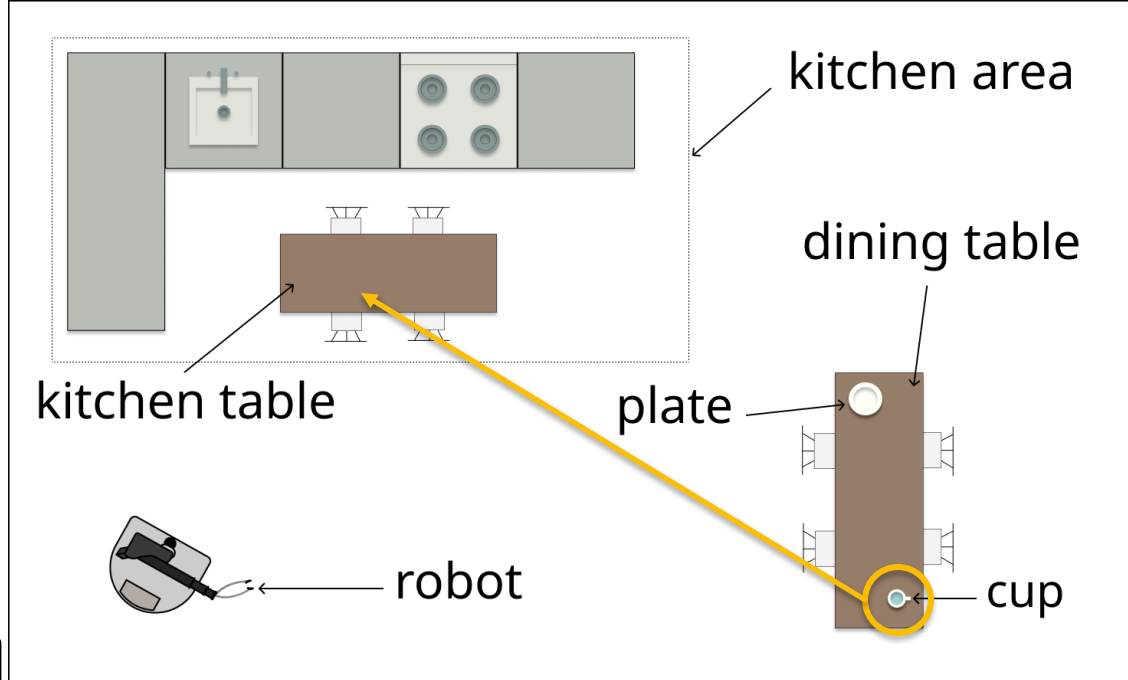
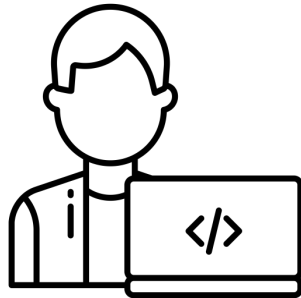
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Problem

- **Task**: Get the cup on the kitchen table.
- Should robot tasks be specified in terms of **goals** or **actions**?



What is the best way to express tasks for a robot?

GOALS

1

The cup is in
your gripper.

2

The cup is
on the
kitchen table.

ACTIONS

1

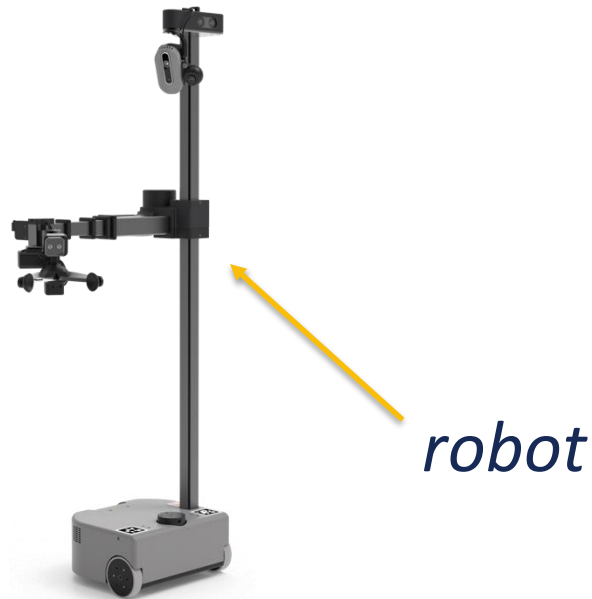
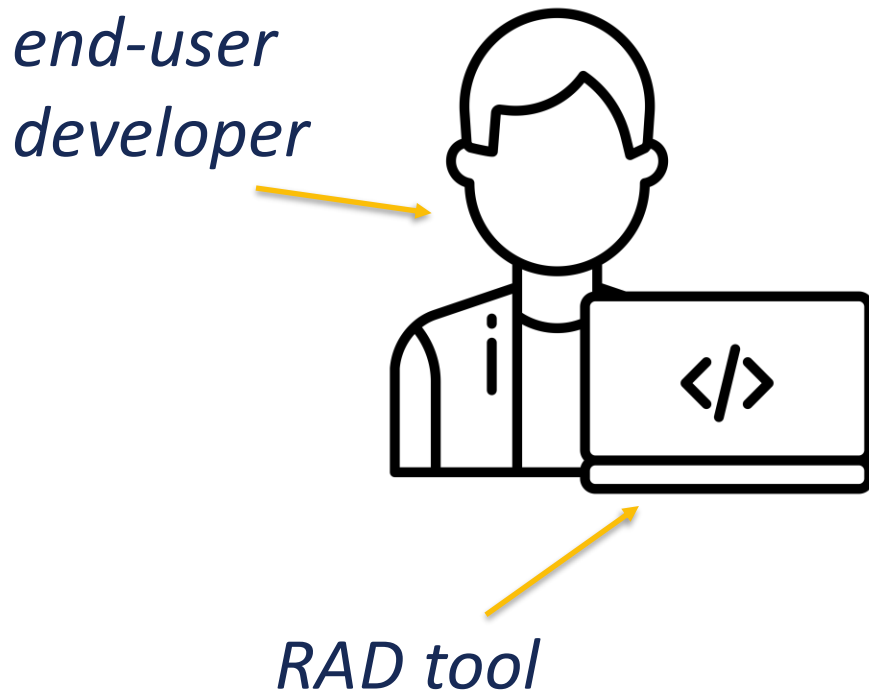
Pick up
the cup.

2

Put the cup
on the
kitchen table.

1. Background and Related Work
2. Research Question
3. Research Plan

Robot application development (RAD) tools



(2024). Stretch 3 [Photograph]. hello-robot.com/stretch-3-product

GOAL

**cup on
kitchen
table**

ACTION

**put cup
on kitchen
table**

Prior Work: Action-Oriented Paradigms

- **Visual Programming Environments:** Interaction Composer (Glas, Kanda, and Ishiguro 2016), Google Blockly (Gargioni and Fogli 2024), Roboflow (Alexandrova, Tatlock, and Cakmak 2015)
- **Mixed-Reality:** V.RA (Cao et al. 2019b), GhostAR (Cao et al. 2019a)
- **Natural Language Programming:** Chat interfaces (Beschi, Fogli, and Tampalini 2019), Spoken languages (Forbes et al. 2015)

Prior Work: Goal-Oriented Paradigm

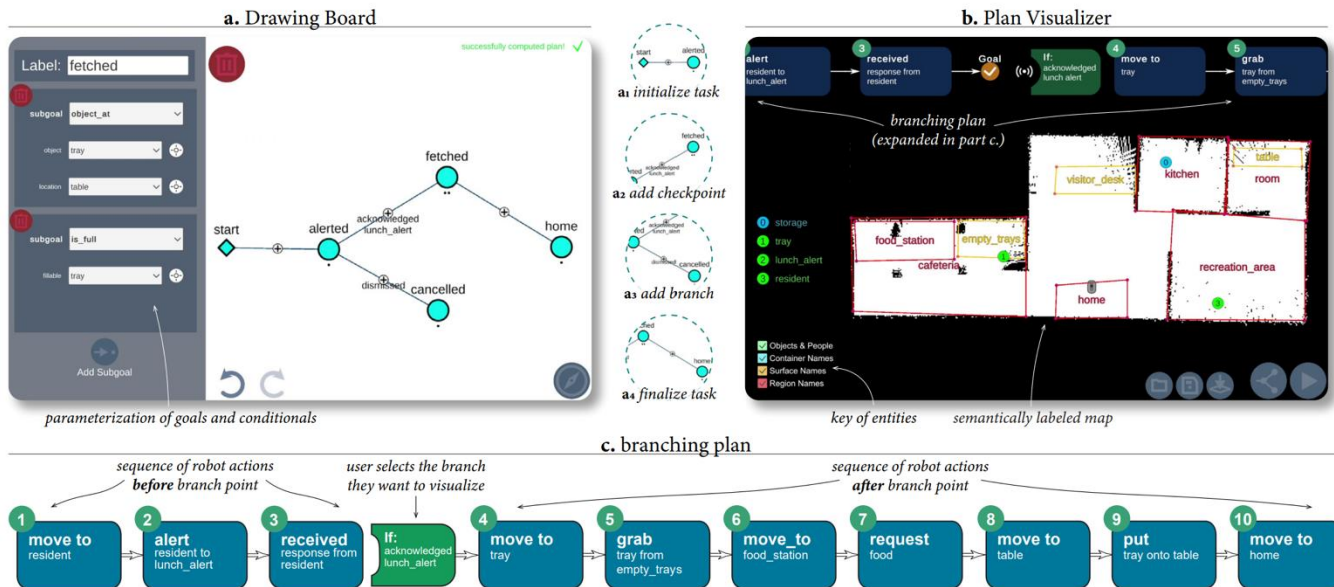


Figure 2: The *Polaris* user interface, which includes (a) the *Drawing Board* for specifying goal automata and (b) the *Plan Visualizer* for displaying the robot's plan. (c) A single branch of the branching plan from the caregiving scenario is displayed.

Prior Work: Human Bias in Reasoning

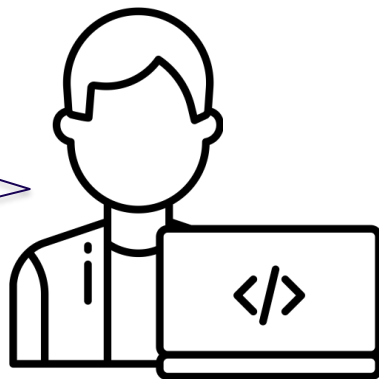
Hypothesis: *Actions* will be strongly preferred over *goals* based on prior work from Forward and Backward reasoning. (Trafton and Reiser 1991)

1. Background and Related Work
- 2. Research Question**
3. Research Plan

Research Question

How should end-user intent be represented as a computational artifact in RAD tools?

Should I specify tasks to the robot in the form of **goals, actions, or both goals and actions?**



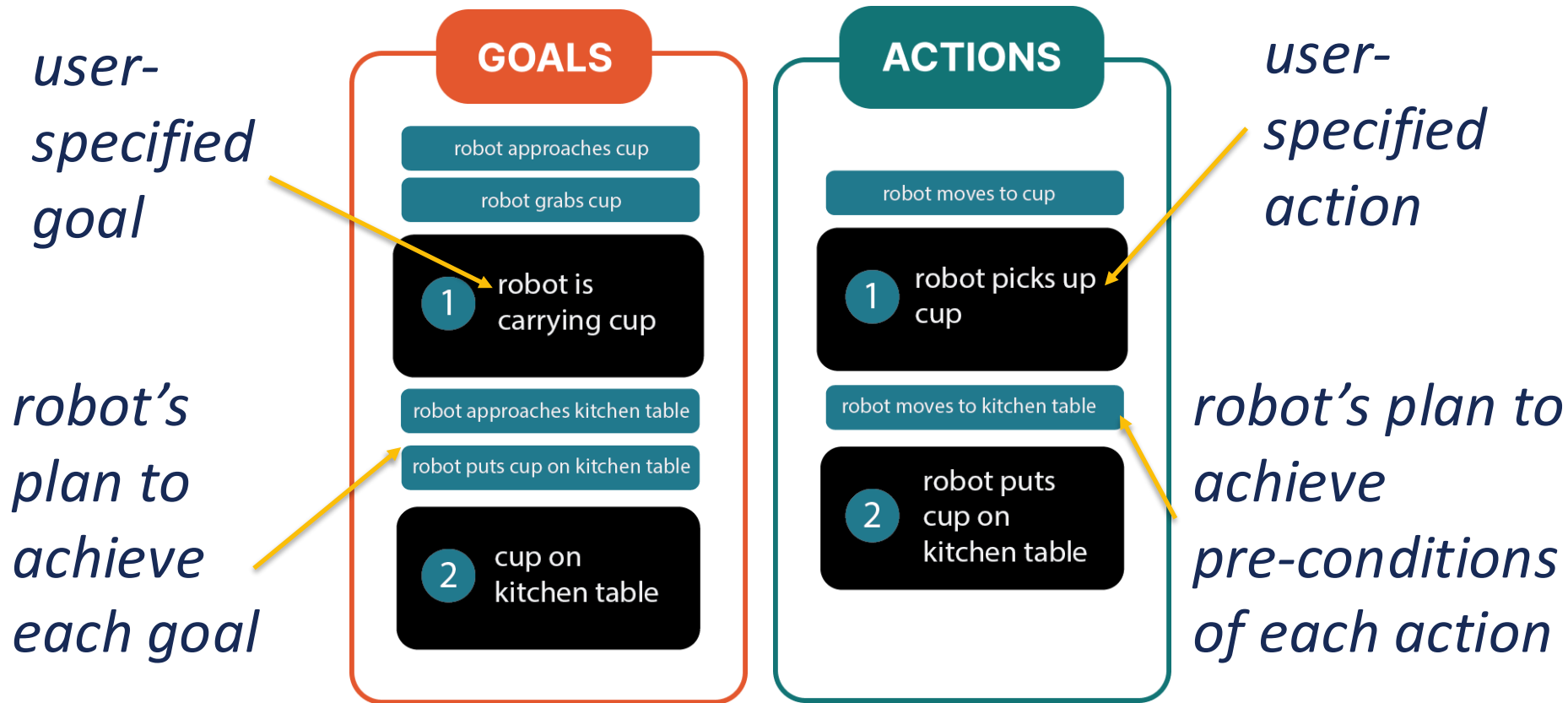
(2024). Stretch 3 [Photograph].
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1. Background and Related Work
2. Research Question
- 3. Research Plan**

Research Plan

1. User Interface
2. Study Design
3. Measures

User Interface



Study Design

- Within-subjects study
- 3 study conditions:
 1. Goals
 2. Actions
 3. Both goals and actions

Measures

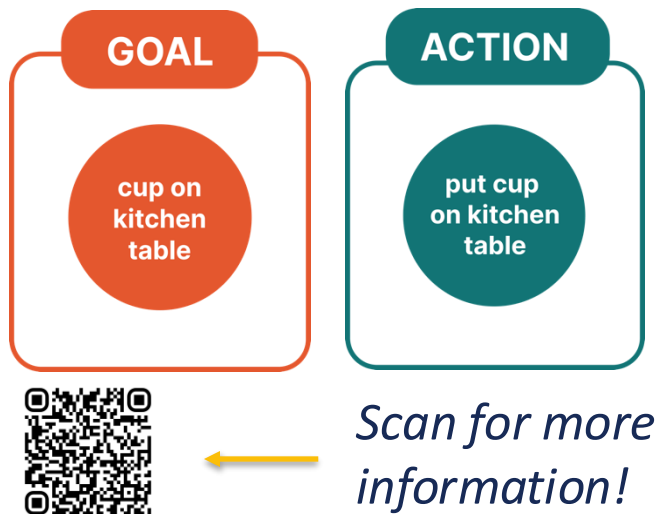
- **Robot perceptions:** trust, expectations, usability, cognitive load, perceived agency
- **Robot performance:** objective reliability, efficiency, time to complete tasks
- Which representation did participants prefer in the *both* condition?

Goals vs. Actions as User-Facing Representations for Robot Programming

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- What is the best way for users to express tasks for a robot?
- Should user-specified tasks be in the form of **goals**, **actions**, or **both goals and actions**?



Acknowledgements: I want to thank Dr. David Porfirio, Dr. Laura Hiatt, and Dr. Mak Roberts from the Navy Center for Applied Research in Artificial Intelligence for their mentorship!